

Honors Calculus III, Homework 7

Please upload solutions to **Questions 1, 2, 3** by **8pm on Monday, May 15**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says “optional”. As usual, for full credit you need to show and explain your work.

1. (5pt each) Let $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ be a function and let $x \in \mathbf{R}^n$.
 - (a) Assume that f is constant. Prove from the definition that f is differentiable at x and $Df(x) = 0$.
 - (b) Assume that f is linear. Prove from the definition that f is differentiable at x and $Df(x) = f$.
2. (5pt) Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be a differentiable function, and let $x, v \in \mathbf{R}^n$. Prove that the rate of change of f at x in the direction v is equal to $Df(x)(v)$. (Hint: use the chain rule.)
3. (5pt) Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be differentiable at $\alpha \in \mathbf{R}^n$. Assume that $f(\alpha) \neq 0$. Prove that $1/f$ is differentiable at α and that

$$D(1/f)(\alpha) = -\frac{1}{f(\alpha)^2} Df(\alpha).$$

4. Calculate the matrix of the derivative at (x, y, z) of the following functions.
 - (a) $f(x, y, z) = x^2 + y^3 + z^5$.
 - (b) $f(x, y, z) = (\sin(x^2 + y), \exp(x + y + z))$.
 - (c) $f(x, y, z) = (x + y + z, x - y - z, x + 2y + 3z)$.
 - (d) $f(x, y, z) = xz \log(1 + y^2)$.
5. Problems from Spivak, *Calculus on manifolds*: 2-10, 2-22, 2-28, 2-34.